

Modeling and optimization of multi-energy systems in mixed-use districts: A review of existing methods and approaches

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ABSTRACT

About 75% of the world's energy consumption takes place in cities. Although their large energy consumption attracts a large number of research projects, only a small fraction of them deal with approaches to model energy systems of city districts. These are particularly complex due to the existence of multiple energy sectors (multi-energy systems, MES), different consumption sectors (mixed-use), and different stakeholders who have many different interests.

This contribution is a review of the characteristics of energy system models and existing modeling tools. It evaluates current studies and identifies typical characteristics of models designed to optimize MES in mixed-use districts. These models operate at a temporal resolution of at least 1 h, follow either bottom-up or hybrid analytical approaches and make use of mixed-integer programming, linear or dynamic.

These characteristics were then used to analyze minimum requirements for existing modeling tools. Thirteen of 145 tools included in the study turned out to be suitable for optimizing MES in mixed-use districts. Other tools where either created for other fields of application (12), do not include any methodology of optimization (39), are not suitable to cover city districts as a geographical domain (44), do not include enough energy or demand sectors (20), or operate at a too coarse temporal resolution (17). If additional requirements are imposed, e.g. the applicability of non-financial assessment criteria and open source availability, only two tools remain.

Overall it can be stated that there are very few modeling tools suitable for the optimization of MES in mixed-use districts.

1. Introduction

About 75% of the world's energy consumption takes place in cities, which in turn causes 70% of worldwide carbon dioxide emissions [1]. These numbers will likely increase with a projected doubling of the urban population by 2050 [2]. To reduce the energy consumption associated with this growth, it is mandatory to increase the energy efficiency of urban systems. While the field has attracted significant research on the potential of reduction of energy use [3], only 14% of the documented research projects have addressed the subject of energy efficiency at the district level [4]. This contribution provides an overview of the methods and tools used to model the optimization of energy systems at urban district levels as these are manageable urban planning units.

An energy system is defined as a “combined process of acquiring and using energy in a given society or economy” [5]. The focus of this contribution lies on multi-energy systems (MES) of mixed-use districts, i.

e. on the acquisition and consumption of various forms of secondary energy (e.g. electricity, heating, cooling) in urban districts characterized by diverse purposes (e.g. residential buildings alongside industry and/or agriculture).

Models are an essential tool for planning and operating energy systems. In this context, a model may be understood as a simplified representation of a real world's energy system [6]. While models may be developed in many different ways, we focus on mathematical and coded models [3]. In addition to the term “model”, “model generator” and “modeling framework” are also frequently used in the modeling landscape. Although these terms seem to have similar meanings, it is important to make precise distinctions. Model generators are tools that can create models with certain predefined analytical and mathematical properties [7]. The use of model generators can save time. However, due to these predefined properties, there are also limitations with respect to the flexibility of the models they generate. Modeling frameworks are structured toolboxes that include several model generators and specific sub-models [7]. Due to the wider range of tools provided with them,

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List of acronyms

C	capital costs
CEA	City Energy Analyst
CED	cumulated energy demand
Cont	continental
Conv	conventional generation
D	disposal
DP	dynamic programming
E	energy
El	electricity
EE	energy efficiency
Em	emissions
EPT	energy payback time
F	fossil resources
Fi	financial
GIS	geographical information system
H	heat
Hyd	hydrogen

Int	international
LA	level of autonomy
LCE	levelized costs of energy
LESP	loss of energy supply probability
Loc	local
LP	linear programming
m	mass
MES	multi-energy system
MIP	mixed-integer programming
Nat	national
O	operation
P	production
p	share of energy
Reg	regional
Ren	renewable generation
S	social
Stor	storages
T	time

there are fewer limitations, but their use requires more coding effort than the use of a model generator.

This is the first contribution addressing the question whether any existing tools of energy system modeling are suitable for the optimization of MES in mixed-use city districts. For this purpose, existing modeling tools are identified, minimum requirements for these tools to be applicable are defined, and the tools are evaluated accordingly. The next phase of the review entails discussing general deficits and making recommendations for further research.

2. Review method

To answer the research question formulated in the introduction, it is first necessary to determine and examine the general features and properties of energy system models (Section 3). The first step thus meant researching review articles on energy system models using Google Scholar as well as FINDEX, Münster University of Applied Sciences' library system. Search terms such as "energy system modeling review" were used. Additional articles were uncovered by looking at these articles' citations as well as by using "cited by" functions. The next step was to search additional articles on the specific properties of energy system models by using the same search methods. A total of 33 review articles have been evaluated [8–41].

The following step required describing and elaborating on the specific characteristics of models for the optimization of MES in mixed-use city districts. This in turn meant identifying the relevant articles using the same methods as above, but including search terms such as "energy AND optimization AND (quarter OR district OR city OR neighborhood)". The corpus included studies published between January 2014 and October 2019. In total, 30 studies have been included [42–72].

The second part of the paper (Section 4) presents and evaluates existing modeling tools. Different reviews [12,14,22,36–41] as well as complementary research with the above mentioned search engines were used as data sources. Only tools that were mentioned in peer-reviewed literature were included in the study. For the specification of specific model properties, "grey literature" (websites, documentations of tools, theses) was also used.

Based on the characteristics of models for the optimization of MES in mixed-use districts (Section 3), we defined minimum requirements that the modeling tools had to have for this purpose. These requirements were then used to categorize and evaluate the various tools.

3. Properties of energy system models

Several reviews exist in the area of energy system modeling. They can be categorized into reviews discussing.

- energy system models in general,
- specific properties of energy system models,
- energy system models for a specific scope of application, and
- existing modeling tools.

Reviews discussing **energy system models in general** provide a broad overview of existing modeling approaches and properties of energy system models. For example, Herbst et al. [8], Jebaraj and Iniyar [9] and Subramanian et al. [10] have presented overviews about existing modeling approaches in general. Foley et al. [11] have reviewed electric system models, while Hall and Buckley [12] evaluated and categorized energy system models in Great Britain. Pfenninger et al. [13] as well as Lopion et al. [14] have discussed current challenges in energy system modeling.

Other reviews deal with **specific properties of energy system models**. For instance, Lund et al. [15] discussed differences between the methodologies of simulation and optimization; Weijermars et al. discussed the methodology of optimization [16] (see Section 3.1, "Methodology"); Stein [17] summarized several *assessment criteria* of electricity production technologies; van Vuuren et al. [18] reviewed the differences between the analytical approaches of top-down and bottom-up approaches (see Section 3.4, "Analytical Approach"). Moreover, Kalogirou [19] and Zahraee et al. [20] dealt with the *mathematical approach* of artificial intelligence, and Pfenninger et al. [21] provided an overview of the importance of open data and software in energy research (see Section 3.6, "Reusability").

There have also been various approaches to using **energy system models for a specific scope of application**. One widely discussed topic was the integration of renewables into existing energy systems (Ringkjøb et al. [22], Després et al. [23], Bhandari et al. [24], Olsthoorn et al. [25], Luna-Rubio et al. [26]). Gu et al. [27] and Fathima et al. [28] focused on the optimization of microgrids; Suganthi and Samuel [29] on energy demand forecasting; Mancarella [30] on models for multi-energy system (MES) concepts; and Hiremath et al. [31] on models for decentralized energy planning. In the context of this paper, modeling approaches of the energy systems of buildings and residential areas are also of interest. Harish et al. [32] reviewed building energy system models in general, Nguyen et al. [33] looked at different approaches to increasing

the energy efficiency of buildings, and Ma et al. [34] reviewed methods of retrofitting buildings using models. Swan and Ugursal [35] analyzed techniques to model end-use energy consumption in the residential sector, while Keirstead et al. [3] provided a more general overview of the approaches, challenges and opportunities facing urban energy system models.

There are also reviews specifically focusing on **existing modeling tools**. Allegrini et al. [36] analyzed 24 existing modeling tools with a focus on the building sector. Ringkjøb et al. evaluated 75 modeling tools with regard to their suitability for modeling renewable energies [22]. Van Beuzekom et al. [37] discussed 13 optimization and planning tools on their suitability for sustainable urban development; Lopion et al. presented 24 currently used tools for energy system modeling [14]; Hall and Buckley classified 22 tools used in the UK [12]. Bhattacharyya and Timilsina [38] presented six tools with a wide scope of application; Connolly et al. [39] discussed 37 tools for analyzing the integration of renewable energies into various energy systems. Suresh and Meenakumari [40] looked at 17 tools, while Sinha and Chandel [41] presented 19 tools that modeled hybrid renewable energy systems.

These articles show that energy system models could be categorized according to their purpose, methodology, assessment criteria, and structural and technological detail. They also take different analytical and mathematical approaches, and vary according to their reusability. These categories are shown in Fig. 1 together with additional characteristics.

3.1. Methodology

The methodology of energy system models can be classified into the three main categories of.

- optimization,
- forecasting, and
- back-casting (Fig. 2).

Optimization models are used for the purposes of investment or operational decision support. The model simulates all possible scenarios of a system, and rates them according to an objective function. Minimizing or maximizing this function helps the researcher to then identify an “optimum” scenario [13,15]. The optimum scenario could, for instance, consist of a preferred mix of technologies [16] or of certain operation modes [73]. Optimization models are the most common type for district-level energy system modeling. Prieto et al. for instance, used models to analyze and optimize the district heating system of Vienna [42]. Fonseca et al. implemented “City Energy Analyst” (CEA), which is a model framework for the analysis and optimization of energy systems in neighborhoods and city districts [61]. Bakken et al. created a model generator for the quantification of the minimum total energy system costs to meet predefined energy demands of electricity, gas and heating [75].

Forecasting models are used for the purposes of system analysis and scenario analysis. They investigate the behavior of a system under given conditions and attempt to predict the system’s future behavior or specific parameters of the system [15], such as the future energy demand [29]. Forecasting is often referred to as “simulation”. However, since simulations are also components of optimization and back-casting models, this term is misleading. The term “forecasting” will therefore be used instead from here on. Forecasting models are not used very often in district-level applications. However, they are very useful when forecasting the demand for renewable energy. Powell et al. for instance, used forecasting models to predict the heating, cooling and electrical loads of a college campus up to 24 h in advance by using weather data as input variables [65].

Back-casting models are used for specific scenario analysis. The researchers lay out an envisaged future state or set of properties of a system, and the back-casting model develops realistic paths that would

lead to these future conditions [12]. This kind of model comes into use during the planning or implementation of political goals [74]. The utilization of back-casting models is even less relevant for applications at the district level. They are more often used for larger system scales, e.g. for nationwide or worldwide energy systems. The World Energy Council, for example, regularly identifies pathways that policymakers should select to achieve specific goals in the world energy system [76].

Several model types may be combined within a single model. Kampelis et al. for instance, implemented a tool to predict the energy demand at the levels of individual buildings as well as district (forecasting) and which would customize the operation of available micro power plants and storage facilities to minimize the energy purchased from external sources (optimization). When they applied their tool to an industrial district in Italy, they found cost savings of up to 15.39% [51].

3.2. Assessment criteria

Identifying the “best” future scenario by optimizing MES in models of mixed-use districts means selecting and applying different assessment criteria. These may be categorized into.

- Financial,
- Energy efficiency,
- Environmental,
- Technical, and
- Social/economic criteria.

Financial criteria aim to identify the least expensive scenario. Useful criteria could include the annualized capital costs C_{cap_a} , the annual system costs C_a , the payback time, the annual net profit, or the levelized costs of energy LCE .

The annualized capital costs C_{cap_a} may be calculated using an estimated rate of interest i and an estimated service lifetime of the system components Y in years [28]:

$$C_{cap_a} = C_{cap} \frac{i(i+1)^Y}{(i+1)^Y - 1}$$

Next, the annual system costs can be calculated from the annual capital costs C_{cap_a} plus additional costs for operation (including maintenance) C_{O_a} [28]:

$$C_a = C_{cap_a} + C_{O_a}$$

The levelized costs of energy (LCE) are calculable from the annualized system costs C_a and the energy delivered over the same period E_a [77]:

$$LCE = \frac{C_a}{E_a}$$

Additional financial criteria could include total life cycle costs or total capital costs [17].

Energy efficiency criteria may measure energy balances such as primary energy use, secondary energy use or consumer (end-user) energy use [53]. Alternatively, criteria measuring the level of autonomy LA of the system or the loss of energy supply probability $LESP$ [26,28] can aid in measuring the degree of self-sufficiency in a system.

We can study energy balances by determining cumulated energy demand CED , which in turn can be calculated by looking at the total value chain of an energy system’s components. This means calculating the energy demands of production (P), of operation (O) (including maintenance), and of disposal (D) after the operational lifetime of the system [79]:

$$CED = CED_P + CED_O + CED_D$$

The energy payback period EPP describes how long it would take to break even on the cumulated energy demand CED [80]:

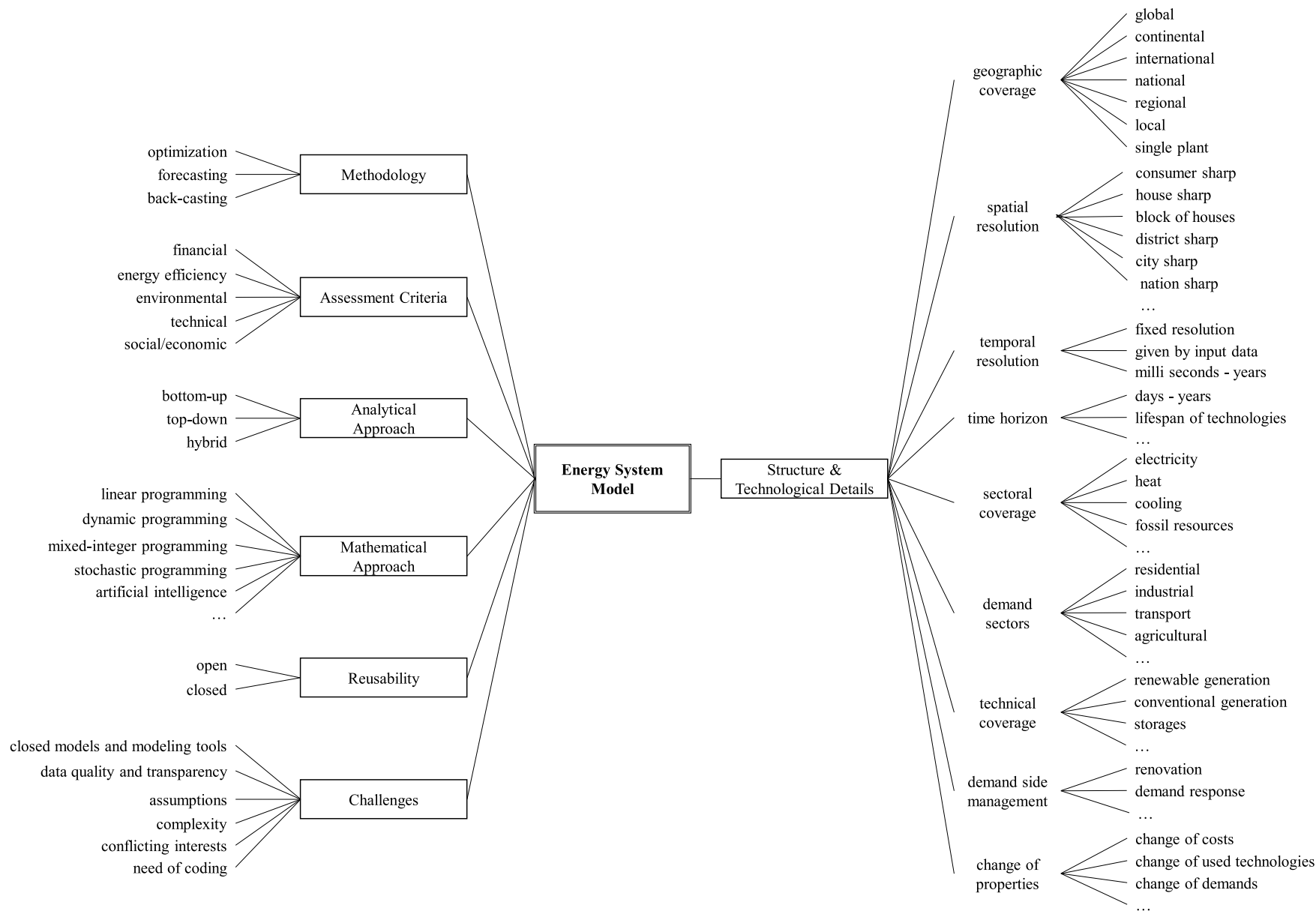


Fig. 1. Properties and possible categorizations of energy system models.

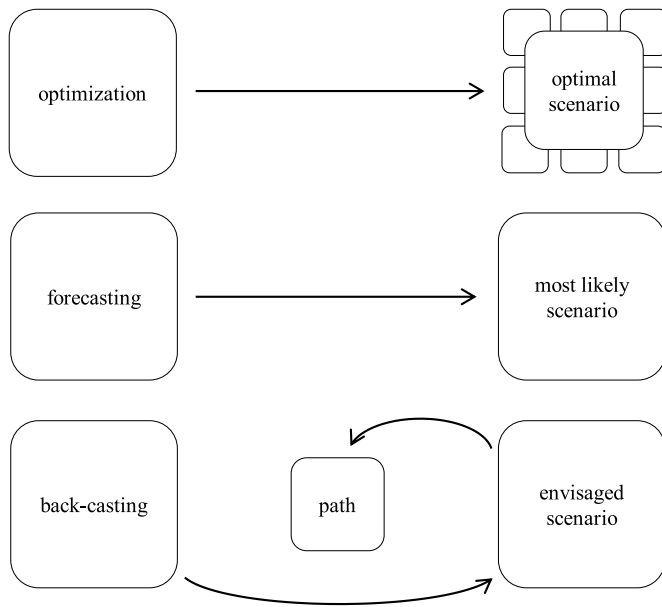


Fig. 2. Methodologies of optimization, forecasting and back-casting.

$$EPP = \frac{CED}{E_a}$$

The level of autonomy LA may be understood as the extent to which an energy system can be operated independently from external or upstream energy systems [26,81]. It can be calculated using the time in which loss of load occurs T_{LOL} and the total operation time T_{tot} [81]:

$$LA = 1 - \frac{T_{LOL}}{T_{tot}}$$

The loss of energy supply probability $LESP$ is the share of energy that cannot be provided by the energy system itself [26] and which has to be obtained from external energy systems. It can be calculated by dividing the sum of energy deficits $E_{def}(t)$ by the total energy demands $E_{dem}(t)$ at any point in time t [26]:

$$LESP = \frac{\sum E_{def}(t) \cdot \Delta t}{\sum E_{dem}(t) \cdot \Delta t}$$

Possible **environmental criteria** could include the amount of greenhouse gases emitted, air pollutants such as NO_x or SO_x , the amount of freshwater used [22], land use, noise pollution, or human health impacts [17]. The amount of CO_2 emitted, m_{CO_2} , for instance, may be calculated throughout the value chain of an energy system's components. Emissions issued by production (P), operation (O) (including maintenance) and disposal (D) after decommissioning all need to be included [82]:

$$m_{CO_2} = m_{CO_{2P}} + m_{CO_{2O}} + m_{CO_{2D}}$$

Another useful environmental criterion would be the percentage of renewables. This is calculated using the percentages of energy supplied by renewable E_{ren} and conventional energy systems E_{conv} :

$$p_{ren} = \frac{E_{ren}}{E_{ren} + E_{conv}}$$

Other environmental criteria, such as health effects, are far less straightforward to quantify, and are thus less frequently applied in system analysis [17].

Technical criteria consider the safety, availability, reliability or overall technical feasibility in a system [17]. Since the processes and components entered to a model usually have to meet state-of-the-art standards of safety, availability and feasibility anyway, there is rarely any need to apply additional technical assessment criteria during the

modeling process.

Social and economic criteria might include regional employment rates or technology-specific job opportunities, the risk of accidents, the availability of resources, the impact on local development and living standards, or foreign trade balances [17].

Financial criteria are probably the most common assessment criteria used to optimize MES at the district level. Environmental and energy efficiency criteria are also often integrated into the modeling. This is especially true of the amount of CO_2 emitted and the percentage of renewables.

Multi-objective optimization approaches – taking more than one criterion into consideration – are frequently used for the modeling of MES at the district level. Orehounig et al. [64], for instance, used the amount of CO_2 emissions (environmental criterion) and the end energy efficiency of buildings, as well as the degree of energy autonomy (energy efficiency criteria) to optimize the integration of decentralized energy systems into neighborhoods. One modeling framework, CEA, allows researchers to integrate financial criteria (levelized energy costs and total annual costs), energy efficiency criteria (amount of primary energy used), environmental criteria (greenhouse gas emissions) as well as social/economic criteria (percentage of renewables) [61].

3.3. Structure and technological details

The structure and technological details of energy system models can be summed up by their.

- Geographic coverage,
- Spatial resolution,
- Temporal resolution,
- Time horizon,
- Sectoral coverage,
- Demand sectors,
- Technical coverage,
- Demand-side management, and
- Change of properties,

as well as other additional properties.

Geographic coverage describes the spatial area that is included in the model. Coverage can range from the global to the continental, international, national, regional and local or even down to a single power plant [22]. The analysis of city districts requires local coverage. A recent model by Spielmann et al. for instance, identified the optimal energy supply scenario of a university campus [53].

The **spatial resolution** of the models could vary as well. For instance, it is possible to map the energy demand of a city district for the whole district, for every building, or even for every consumer. The **temporal resolution** specifies the time steps of the model. This could range from milliseconds to years (Table 1 [11]). The actual temporal resolution is fixed in some models and variable in others, for example depending on the resolution of the input data [22]. Ringkjøb et al. stressed that energy models with broad geographic coverage tend to have lower temporal resolution [22]. Models at the district level generally use an hourly resolution, less often a 15-min resolution. One-minute resolutions are used as well, but very rarely. Depending on the specific research question, more detailed results may be modeled by applying higher spatial and temporal resolution. However, the computing resources required increases quickly as spatial and temporal resolution of the model increases. Moreover, input data is usually not available at such high levels of resolution, so a better resolution of the model itself may not lead to better results [13].

Examples of the **time horizon** might include a day, a year, a decade, or the lifespan of the components used in the energy system [12]. Optimization models at the district level typically cover at least one full year.

Sectoral coverage describes which energy sectors are included in

Table 1
Applications of different temporal resolutions.^a

Temporal Resolution	System Issues	Applications
ms → s	- Generator dynamics - Motor load dynamics	- Stability management - Power-frequency regulation
min → 1 h	- Demand variation - Power interchanges - Maintain economic operation - Frequent control - Hourly generation planning	- Economic dispatch - Generation control - Power flow - Security analysis - Fault analysis - Voltage stability studies
h → 1 week	- Weekly generation planning	- Demand - Weather prediction - Unit commitment
weeks → months	- Seasonal generation planning	- Demand prediction - Maintenance planning - Hydro planning - Fuel planning
Years	- Demand growth - Plant Retirement/refurbishment - Investment opportunities - Long-term hydrological cycles	- Generation expansion planning - Reliability checks - Scenario analysis - Production cost modeling

^a Reprinted from Energy, 35/12, A.M. Foley, B.P.Ó. Gallachóir, J. Hur, R. Baldick, E.J. McKeogh, A strategic review of electricity systems models, pp. 4522–30, Copyright (2010), with permission from Elsevier [11].

the model. These energy sectors typically include electricity, heating (district heating, low exergy networks, hot water), cooling, fossil resources, and commodities.

Possible **demand sectors** include the residential and commercial building sectors, industry, agriculture [22] and transport [12]. Models of mixed-use districts should incorporate at least two of these sectors.

Technical coverage refers to the specific technologies applied. Optimization models at the district level generally take into consideration solutions such as decentralized energy supply from combined heat and power plants (CHP), the renewable technologies of solar thermal plants, photovoltaics and heat pumps, and various types of heat and electricity storage methods, as well as boilers for heat supply. Occasionally, other technologies such as small hydropower plants [64], wind turbines [55,66], biomass boilers [54] and diesel engines [28] are also incorporated into models.

Demand side management comprises all measures on the consumer side of an energy system. They are typically intended to increase the energy efficiency of the system [83]. Measures may range from renovation (e.g. the installation of well-insulated windows) to demand-response measures (such as shifting energy demands from periods with high demand to periods with energy surpluses) [22]. Such measures can be especially helpful to balance the volatile production rates of renewable energy systems. They may be supplemented by energy storage facilities [22]. Only a very few studies have incorporated renovation measures into optimization models at the district level. One exception was a study by Falke et al. which dealt with the question of whether renovation measures in buildings could be an alternative to investment in new generation units [59]. Demand response has also taken on new importance in recent years, especially with the emerging application of artificial intelligence (see the section on *mathematical approaches below*). Kampelis et al. [51], for instance, developed an optimization approach for buildings and districts using demand-response measures in connection with artificial intelligence, and came to the conclusion that such measures could lead to cost savings

of up to 15%.

Especially in long-term analyses, some of the **properties** of an energy system may **change** over the defined time horizon. In addition to changes in costs and components used, changes in behavior among residents could have profound effects on the demand side. Changes in system properties can be included in optimization models if the models include some kind of forecasting abilities. In current models, changes in system properties is only considered occasionally.

Additional properties of a model may, for instance, provide information about various cost aspects in the model [22] or how uncertainty and risk is treated [12].

3.4. Analytical Approaches

We can classify the analytical approaches that models of energy systems can take into three categories:

- Bottom-up models,
- Top-down models, and
- Hybrid models.

The first two are depicted in Fig. 3. In a **bottom-up** approach, several subsystems are modeled, and then these are merged onto an overlying system. Assembling the whole system from the bottom to the top makes it possible to create a model with a high level of technological detail, which allows for the evaluation of a wide range of technical options [84]. However, investment and cost aspects are usually limited to the technologies specifically included in the model. This leads to a lack of attention to macroeconomic interactions between the energy sector and other economic sectors. For example, changes in consumer behavior due to cost changes are hard to estimate [18]. Bottom-up energy system models are mainly used to identify technologies suitable for certain applications [84]. They are traditionally employed in the field of engineering [85], and are also known as techno-economic models [8].

The process of building a model with a **top-down** approach starts with a general overview of the entire system, and becomes more detailed by splitting the whole system into subsystems. Such models usually include the whole economy but take into account fewer technological details [18]. To compensate for this lack of detail, models often use historical parameters to estimate future system behaviors, which may cause imprecise results [84]. This approach is traditionally used for economic investigations (e.g. for the prediction of future energy demands, or conditional on macroeconomic interactions [8,18]) and is also known as the macroeconomic approach [8].

Hybrid models are created by linking a top-down model with a bottom-up one, combining the advantages of both approaches [8,12]. Despite the additional effort needed to apply this approach, more and more energy system models are based on hybrid approaches [8,18].

For the technical optimization of the holistic MES of urban districts, a modeling with high technical detail is mandatory. Such technical detail can only be guaranteed if the technical properties of various components are mapped in detail and then combined into an overall system. Therefore, models usually employ bottom-up approaches, and less often hybrid approaches. Top-down models are normally not applied for this purpose. Petrovic and Karlsson [57], for instance, implemented a model with the TIMES model generator, using the bottom-up approach. They investigated the role of heat pumps in residential energy systems to achieve Denmark's energy policy targets. They found that heat pumps in Denmark could supply 24–28% of total heat demand. On the other hand, they also determined that the energy targets could be met without using heat pumps at all. This would, however, increase total costs by 16%.

3.5. Mathematical Approaches

Different models take different mathematical or rather computational approaches as well. The most common include the following:

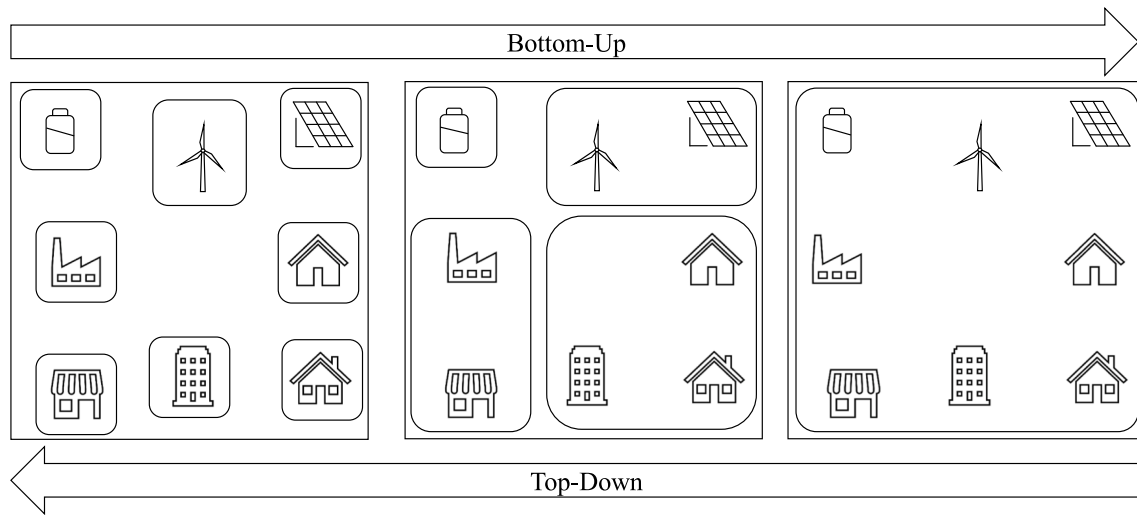


Fig. 3. Principle of bottom-up and top-down approaches.

- Linear programming,
- Dynamic programming,
- Mixed-integer programming,
- Stochastic programming, and
- Artificial intelligence approaches.

With the use of **linear programming** (Fig. 4 a), system behavior is described by linear mathematical functions, which in turn are summarized in an objective function. The model result is reached, for example, by finding and identifying the minimum or maximum of the objective

function. Linear optimization models have a long history in energy system modeling [13,86], and are frequently used for the optimization of MES at the district level. Linear programming approaches are straightforward to apply. However, they are only appropriate in cases when all system relations can be thoroughly described with linear functions [87].

Non-linear relations either have to be linearized (Shao et al. for instance, transformed dynamic energy demands into piecewise-linear functions [88]) or instead treated with **dynamic programming** (Fig. 4 c). This is an algorithmic approach to solving problems by

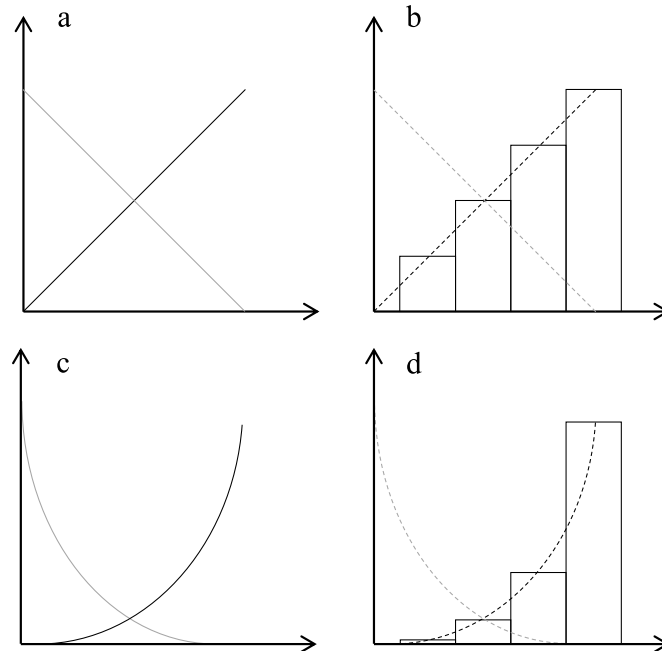


Fig. 4. Linear programming (a), linear mixed-integer programming (b), dynamic programming (c) and dynamic mixed-integer programming (d).

splitting them into sub-problems and systematically storing intermediate results. The subsequent combination of the intermediate results yields an overall solution [89].

Mixed-integer programming (MIP, Fig. 4 b, d) is a particular type of the three approaches mentioned above, but only allows outcomes with integer values [22,87]. These are useful in cases when it is helpful to identify an integer number, such as the ideal number of power plants, or storage facilities with predefined capacities.

The differences between linear and dynamic programming, as well as the differences between MIP and non-MIP are illustrated with a simplified example in Fig. 4. For example, the target values could be the point of intersection of the functions, or the minimum or maximum of the sum of the functions. Depending on the type of programming, these points would be different. For MIP functions, there is a limited number of solutions (represented by blocks), while for non-MIP functions, an infinite number of solutions is theoretically possible.

Stochastic programming approaches are linear and dynamic programming approaches that are not limited to the use of fixed parameter sets. Imprecisions in parameters and functions are coded into probabilistic elements, which in turn allow the generation of uncertainty estimates of the model output [11].

Artificial intelligence approaches are even further advanced and more complex approaches. Here, not only the imprecision of properties, but even a change of properties within a given time horizon is detected or predicted by the model itself. A decisive advantage of artificial intelligence is that it is fault-tolerant and can therefore also work with noisy or incomplete data. In addition, it is capable of working faster than conventional algorithms [19]. In addition, the results obtained from artificial intelligence can contribute to deeper understanding of the system. The implementation of artificial intelligence, however, is much more complicated than that of conventional optimization algorithms, and requires in-depth programming knowledge, especially as the number of parameters increases [20]. An additional challenge is that previous solved scenarios must be available and known to train the artificial intelligence [20,24]. Artificial intelligence approaches are based on fuzzy logic [12,24], agent-based programming [12], particle swarm optimization [20], genetic algorithms, or neural networks [24].

Mixed-integer programming is the approach most commonly used for the optimization of MES at the district level, while dynamic approaches are used less frequently. Artificial intelligence has only been used quite rarely so far, but has begun to receive more and more attention recently. Stochastic programming is used occasionally. Morvaj et al. [58], for instance, implemented a model using linear MIP to optimize district heating layouts. They identified measures to reduce the system's CO₂ emissions by 23% without any increase in costs. Stochastic models can also be used to determine how reliable an energy system would be under emergency conditions, as Najafi et al. [90] have done. Reynolds et al. [43] used artificial neural networks and genetic algorithms to predict multiple variables and to optimize operating schedules of heat generation units, thermal storage units, and heating set points. The optimization would allow a 45% increase in profit compared to a rule-based, priority-order baseline strategy. Schwarz et al. [45] used a stochastic linear MIP approach to support investment decision-making for technologies with unstable energy generation (photovoltaics and heat pumps) in a residential area. They predicted the required capacities of generation and storage to ensure reliable power supply under various probability and risk scenarios.

3.6. Reusability

Depending on their availability and reusability, it can be differentiated between models and modeling tools (model generators and frameworks) that are.

- Open and
- Closed [7].

Open models and modeling tools are characterized by the public accessibility of their source code, their underlying assumptions, and the data they use [7]. **Closed** models and modeling tools, on the other hand, do not publish their code or underlying assumptions [7]. Their use may be subject to fees, or even completely confidential [7]. The large majority of energy system models, model generators, and frameworks are closed. In light of calls for greater transparency and reusability of publicly funded research in Europe [90], there has been an increasing number of open models and modeling tools over the past few years [21].

In addition to motivations based on market strategies and economic considerations to keep commercially developed models and modeling tools closed, Pfenninger et al. [21] mentioned four reasons why models and modeling tools often remain closed. (1) Models and modeling tools may contain sensitive commercial data or personal information, which is not permitted for public disclosure. (2) The release of a code carries a risk that other researchers could expose flawed code sections or erroneous data and thus discredit the results. (3) The writing of legible and reusable codes as well as comprehensible documentation and bug reports is time-consuming. Not everyone is willing to invest this time. (4) The hesitation among individuals or institutions may also be a cause for failing to open models and their results.

Pfenninger et al. [21] also put forth four reasons why models and modeling tools should be published openly. (1) The fundamental scientific principles of transparency, peer review, reproducibility and traceability can only be guaranteed if data, methodology and results are openly accessible [21,92]. (2) Policymakers often have to fall back on models that are not quality-assured with academic practice or that provide incorrect results. With increasing transparency in energy system research, policymakers will gain access to more high-quality information [21]. (3) Research funding and researchers' time are limited resources. A great deal of time and money can be saved by avoiding duplication of work [21]. (4) The transparency of arguments based on scientific justifications are necessary in social and political debates [93]. Furthermore, the full results of publicly funded research should be available to the public.

3.7. Challenges

There seem to be six major challenges in the field of energy system modeling at present:

- Closed models and modeling tools,
- Data quality and transparency,
- Unreliable assumptions,
- Complexity,
- Conflicting interests, and
- The need for coding.

The high share of **closed models and modeling tools** generally hinders progress in energy system modeling. Data with high **quality** and with a sufficient temporal resolution are indispensable for the production of high-quality model results. Often, data to feed the models is not available, and often, the quality of the data is poor. Even in cases when data meeting the requirements exists, a lack of **transparency** and **accessibility** (arising for various reasons) are real problems [13]. Moreover, models are often based on **assumptions** that turn out to be more or less uncertain. Dealing with this uncertainty can be a challenge. Barely predictable power production from renewable energy systems [14] as well as human behavior, especially feedback between consumer behavior and policy decisions, introduce significant uncertainty into model codes and results [13].

The **complexity** of energy system models is increasing due to increasing geographic coverage, temporal resolution, and time horizons, as well as increasing numbers of relevant economic sectors and technologies. This in turn results in a drastic increase in the effort as well as the computing resources required from the modeler. To limit the need

for those resources at acceptable levels, researchers need to strike compromises between the time and energy put into a model, the computing time necessary, and the informative value of the results [94].

Different stakeholders of energy system modeling may also have **conflicting interests**, which cannot be met within a single model. Therefore, models often represent tradeoffs, trying to combine various objectives, but yielding non-optimal results for any single stakeholder [16]. Policymakers could, for instance, aim to reduce costs (financial criteria), while local residents might instead focus on social or environmental criteria.

Energy system models are usually **coded** models, which means that even if researchers are working with model generators or frameworks, they can only take full advantage of energy system models if they have coding skills. To bring energy system models into widespread use among the public, it will be crucial to develop intuitive operable application surfaces (e.g. graphical user interfaces or spreadsheet/GIS (geographical information system) based applications).

All of these problems and challenges are relevant for the optimization of MES in mixed-use districts. Specific problems could, for instance, include a lack of high-quality load profiles, unreliable assumptions about future consumer behavior, increased complexity due to the inclusion of several economic sectors in the MES, or contradictory requirements for assessment criteria from policymakers or residents. The need for coding is also an obstacle to getting public officials to use models or modeling tools.

4. Existing modeling tools

The analysis of several reviews that discussed existing modeling tools [12,14,22,36–41] as well as an additional online search revealed a total of 145 energy system modeling tools. Not all of them were suitable for the optimization of MES in mixed-use districts. Based on our hitherto analysis and applying careful evaluation, we applied a cascade of criteria, to filter out the relatively small number of suitable modeling tools (Fig. 5):

- 1) In general, the **area of application** of a given tool has to be an energy system as it was defined at the beginning of this paper.

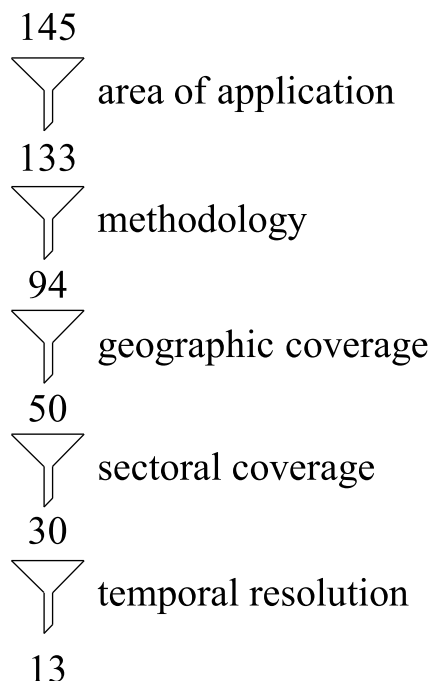


Fig. 5. Filtering of energy system modeling tools by applying the requirement for their suitability of optimizing multi-energy systems in mixed-use districts.

Furthermore, the tool has to be useable for more than one such energy system.

- 2) The applied **methodology** has to include some kind of optimization.
- 3) A city district needs to be **geographically coverable**.
- 4) For the analysis of MES, **sectoral coverage** should include at least the electricity and heat sectors. To take mixed use into account, at least two different demand sectors have to be included.
- 5) The **temporal resolution** must generate at least hourly time steps.

Fig. 5 shows how many tools were filtered out in each step. After applying all criteria, only 13 modeling tools were suitable for optimizing MES in mixed-use districts. These are listed in Table 2 with some of their properties. Table 3 lists the tools we filtered out, together with the criteria that lead to their exclusion.

In total, 132 of the 145 modeling tools analyzed here were not suitable for the optimization of MES in mixed-use districts. A total of 12 tools were designed for another area of application; 39 tools did not include any methodology of optimization; 44 tools were designed for either undersized or oversized geographic coverage (e.g. buildings, nations), 20 tools did not allow the consideration of more than one energy sector or demand sector; and finally, 17 tools had insufficient temporal resolution. Depending on the specific requirements for the envisaged application of a modeling tool, even more optional exclusion criteria may be necessary. If, for instance, assessment criteria other than financial criteria should be incorporated, only four tools are suitable (EnergyPLAN, eTransport, oemof, urbs). Note that five of the 13 suitable models come with an open source license. Two additional models are freely accessible. Six models are commercial, but none of the suitable models are completely inaccessible. Two tools (oemof, urbs) are available with an open source license and, at the same time, integrate various assessment criteria – in addition to financial ones. Overall, there are suitable tools, but only a small number of them. There is an inherent risk in a small number of suitable tools: there are hardly any possibilities to compare model outputs with each other, and thus both errors and potential for improvement are difficult to detect.

The “Open Energy Modeling Framework” (oemof) is a community-operated open-source modeling framework for the analysis and optimization of energy supply systems. It is designed as a modular python library consisting of a set of sub-libraries [123]. Examples of these sub-libraries include “oemof.solph”, which can be used to describe energy systems on the basis of mathematical graph theory, “oemof.demandlib”, which simulates consumers on the basis of standard load profiles, and “oemof.feedinlib”, which enables the modeling of renewable energy systems such as photovoltaic and wind power plants [123]. The community continuously develops new modules. The flexible modeling approach makes it possible to apply any geographical or temporal resolution and coverage; any energy or demand sector, and a variety of assessment criteria.

The modeling tool “urbs” focuses on energy systems that have a high proportion of renewables [189]. It was originally developed for the optimization of urban energy systems, but researchers have also employed it for continent-wide energy systems [189]. The tool was also developed in python and has a modular structure [134]. However, unlike oemof, the application does not require the use of a programming language. The model implementation takes place by entering spreadsheet files [134]. Financial costs and CO₂ emissions can be applied as assessment criteria.

All in all, oemof offers more sub-modules, and its flexible modeling approach makes it more accurate and versatile. On the other hand, urbs has a lower entry hurdle because there is no need to apply programming skills.

5. Conclusion

Energy systems of urban district are mixed-use multi-energy systems (MES). They include several energy sectors, such as electricity and heat,

Table 2

Modeling tools meeting the criteria for optimizing multi-energy systems in mixed-use districts. Acronyms: **Assessment Criteria:** EE = energy efficiency, Em = emissions, Fi. = financial, S = social. **Mathematical Approach:** DP = dynamic programming, LP = linear programming, MIP = mixed integer programming. **Geographic Coverage:** Cont. = continental, Int. = international, Loc. = local, Nat. = national, Reg. = regional. **Sectoral Coverage:** El. = electricity, F = fossil resources, H = heat, Hyd. = hydrogen. **Technical Coverage:** Conv. = conventional generation, Ren. = renewable generation, Stor. = storage (electricity and heat).

Tool	Calliope	DER-CAM	EnergyPLAN	energyPro	eTransport
Methodology	Optimization	optimization	optimization	optimization	optimization, forecasting
Assessment Criteria	Fi	Fi	Fi, EE, Em, S	Fi	Fi.
Analytical Approach	bottom-up	bottom-up	bottom-up	bottom-up	–
Mathematical Approach	LP ^a	MIP	LP	DP	DP, MIP
Geographic Coverage	user defined	buildings and microgrids ^b	Loc. to Cont.	Loc. to Reg	user-defined
Temporal Resolution	user defined	years ^c	hours	minutes	hours
Time Horizon	user defined	max. 20 years	1 year	max. 40 years	1–30 years
Sectoral Coverage	El., H, Hyd., F	El., H, C, F	El., H, F	El., H, C	El., H, C, F
Demand Sectors	any (aggregated)	any (aggregated)	any (aggregated)	any (aggregated)	any (aggregated)
Technical Coverage	Ren., Conv., Stor. ^d	Ren., Conv., ^e Stor.	Ren., Conv., Stor.	Ren., Conv., Stor. ^f	Ren., Conv., Stor. ^g
Demand Response	yes	yes	no	no	no
Accessibility	open source	free	free	commercial	commercial
Website ^h	https://calliope.readthedocs.io/en/stable/index.html	https://building-microgrid.lbl.gov/	https://www.energyplan.eu/	https://www.emd.dk/de/energypro/	https://www.sintef.no/en/projects/etransport/
References	[14,22,95–100]	[12,22,101–103]	[12,14,22,40,104–108]	[12,22,36,107,109–112]	[37,75,113]

Tool	ficus	HOMER	MARKAL	MARKAL-MACRO	oemof
Methodology	optimization	optimization, forecasting	optimization	optimization	optimization
Assessment Criteria	Fi.	Fi.	Fi.	Fi.	Fi., EE, Em, S
Analytical Approach	bottom-up	bottom-up	bottom-up	hybrid	all applicable
Mathematical Approach	LP, MIP	–	LP, DP	DP	LP, MIP
Geographic Coverage	Loc. to Nat.	Loc.	Loc. to Reg.	Loc. to Nat.	user-defined
Temporal Resolution	15 min	minutes	user-defined	user-defined	user-defined
Time Horizon	1 year	several years	user-defined	user-defined	user-defined
Sectoral Coverage	any	any ⁱ	any	any (aggregated)	El., H, Hyd., F
Demand Sectors	any (aggregated)	any (aggregated)	any (aggregated)	any (aggregated)	any (aggregated)
Technical Coverage	Ren., Conv., Stor.	Ren., Conv., Stor. ^j	Ren., Conv., Stor. ^k	Ren., Conv., Stor. ^l	Ren., Conv., Stor.
Demand Response	no	no	yes	yes	yes
Accessibility	open source	commercial	commercial	commercial	open source
Website	https://ficus.readthedocs.io/en/latest/index.html	https://www.homerenergy.com/	https://iea-etsap.org/index.php/etsaptool/model-generators/markal	https://iea-etsap.org/index.php/etsaptool/model-generators/markal	https://oemof.org/
References	[22,114,115]	[12,22,36,40,41,116–118]	[12,14,22,119–122]	[12,119,123]	[7,14,22,124–126]

(continued on next page)

Table 2 (continued)

Tool	Temoa	TIMES	urbs
Methodology	optimization	optimization	optimization
Assessment	Fi.	Fi.	Fi., Em.
Criteria			
Analytical	bottom-up	hybrid, bottom-up	bottom-up
Approach			
Mathematical	LP	LP, DP	LP
Approach			
Geographic	user-defined	user-defined	Loc. to Nat.
Coverage			
Temporal	years ^m	years ⁿ	user-defined ^o
Resolution			
Time Horizon	user-defined	user-defined	user-defined
Sectoral Coverage	any	any	any
Demand Sectors	any (aggregated)	any (aggregated)	any (aggregated)
Technical Coverage	Ren., Conv., Stor.	Ren., Conv., Stor.	Ren., Conv., Stor.
Demand Response	no	–	yes
Accessibility	open source	commercial ^p	open source
Website	https://temoacloud.com/	https://iea-etsap.org/index.php/documentation	https://urbs.readthedocs.io/en/latest/
References	[12,14,22,127–129]	[12,14,22,130–133]	[22,134,135]

^a MIP under development.

^b Districts can only be modeled if they are assessed as microgrids.

^c Temporal resolution of a year, but reference days with minute-level resolution may be defined.

^d Tool designed for questions concerning the transition to renewable energy. The developers have pointed out that there are tools that are more suitable for other types of questions.

^e Excluding nuclear power.

^f Photovoltaic, wind turbines, batteries, electric boilers, electric absorption chillers, thermal storage units, geothermal energy, biomass, biogas and other fuels can be included.

^g CHP, heat pumps, boilers, LNG plants, and power plants without emission flows can be included.

^h Websites last accessed on June 9, 2020.

ⁱ The focus is clearly on the electricity sector.

^j The focus is on renewable generation. Nuclear power cannot be included.

^k Only hydropower, wind power, solar power and geothermal as renewables. Only night-day storage.

^l Only hydropower, wind power, solar power and geothermal as renewables. Only night-day storage.

^m Temporal resolution of multiple years, but a number of reference days for various seasons with undefined time steps may be defined.

ⁿ Temporal resolution of a year, but reference days with hourly resolution may be defined.

^o Usually hours.

^p Open source code, but commercial GAMS license required.

Table 3

Modeling tools that were not suitable for the optimization of multi-energy systems of mixed-use districts.

Exclusion Criterion	Tools		
area of application	BALMOREL [12,14,22,136–139] BESOM [14] Emcas [12,39] HYDROGEMS [12,39] AEOLIUS [12,39] ARES [40,41] AURORAxp [22] CASPOC [22] CIMS [14] COMPOSE [12,22] CYME [22] DESSTinEE [22] DigSILENT/Power Factory [22] Dymola/Modelica [41,94] DynEMo [12,14] EMLab-Generation [22] ETM (2) [22] BCHP Screening Tool [39] CitySim [36,161] EnergyPlus [36,163] ESP-r [36]	Neplan [36,140] NetSim [36,144] ORCED [12,39] PLEXOS [11,12,22,148–150] GridLAB-D [22] H2RES [12,39,152,153] HYBRID 2 [40,41,155] HYBRIDS [41] HybSim [40,41,156] HySys [41] INFORSE [12,39] INSEL [41,157] INVERT/EE-Lab [12,22] IPSA 2 [22] IPSYS [40,41] LOADMATCH [22] MDM-E3 [12] IDA ICE [36] KULeuven IDEAS lib [36] LBNL District lib [36] Polysun [36,164] GEM-E3 [12,22] GENESYS [22] IMAKUS [22] LIBEMOD [22] LIMES-EU [22] LUSYM [22] MiniCAM [12,39] NEMO [22] PERSEUS [12,39] POLES [12,22] RAMSES [12,39] REMIND [22] Hybrid Designer [40,41] HYPERSIM [22] iGRHYSO [40,41] iHOGA [22,40,41,181–183] IRiE [22] NEMS [12] OSEMoSYS [12,14,22] IKARUS [12,14,185–188] LEAP [12,14,22] MESSAGE [12,14,22] MESSAGE-III [12] NEMS [14,22] PRIMES [12,14,22]	ProdRisk [12,22] renpassGIS [21,22,141–143] SynCity [36,146] Termis [36,147] Mesap Planet [12,39,151] MEU [36,154] OPENDSS [22] RAPSim [22] SAM [22] SIMPOW [22] SIREN [22] sivael [12,37,39] SOLSIM [40,41] TRNSYS [41] UKENVI [12] UMI [36,158] WEM [22] SCOPE [14,159,160] TRNSYS18 [22,36,162] REMIX [14,22,165–170] REMod-D [14,171–174] SAGE [12,119] SimRen [12,39] stELMOD [22] Stream [12,37,39] WASP [12] WeSIM [22] WILMAR [12,39] WITCH [22] PowerGAMA [22] PyPSA [22,176–179] ReDS [22] SOLSTOR [40,41] SOMES [40,41] SWITCH [22] REMIND-D [14] RETSCreen [12,22,36,40,41] SNOW [22] TESOM [14] UniSyD3 [12,39]
methodology			
geographic coverage (too small)			
geographic coverage (too big)			
sectoral coverage			
temporal resolution			

and various consumption sectors, such as housing, industry and agriculture. Not least because of the structural change in the European energy system, the energy systems of city districts have become the focus of many planning activities. To meet the resulting challenges, tools for modeling and optimizing the energy systems of urban districts are needed. Although various studies have looked at increasing the energy efficiency of urban districts, or the development of general energy system modeling approaches in general, there is a lack of studies dealing with modeling tools for exactly this purpose.

To be suitable for the optimization of MES of mixed-use districts, modeling tools must be able to apply an optimization methodology. They have to address multiple energy and demand sectors, and operate with an at least hourly temporal resolution. Moreover, they have to follow a bottom-up or hybrid analytical approach. As the analysis presented in this article has shown, only 13 out of a total of 145 modeling tools meet these requirements. The other 132 tools were either created for another area of application (12 tools), used a different methodology (39 tools), were unable to consider the geographic coverage of city districts (44 tools), did not include enough energy and demand sectors (20 tools), or worked with overly low temporal resolution (17 tools).

We are fully aware that a direct comparison of the modeling tools presented here using a representative MES from a mixed-use district would be interesting. All 13 modeling tools meeting the requirements

should be compared to each other. However, Connolly et al. have shown in their review [39] that the individual training time for each modeling tool is usually several weeks or even months. Therefore, carrying out such a comparison is outside the scope of this contribution. If readers are familiar with one or more of these tools, we would welcome the opportunity for a forthright joint study.

Going further and adding optional criteria such as the ability to use non-financial assessment criteria and open source availability, only two of the original 145 tools are suitable. Accordingly, oemof and urbs stand out from the crowd here.

Challenges in the field of energy system modeling, and therefore presumable reasons why the availability of models and modeling tools for urban districts are lacking, include the limited accessibility of supporting data as well as generally poor data quality. Another challenge is the need to find compromises between the preferably large system's complexity on the one hand, and the resources needed for coding and computing on the other. Conflicting interests among various stakeholders often produce difficulties as well. In our view, the lack of transparency is an avoidable problem. Existing databases and modeling approaches should be kept open. This would save time and effort, and eventually lead to a higher quality of energy system models. Fortunately, a significant share of the models analyzed in this review are available with an open source license or are at least freely accessible.

One problem in the literature on the topic of energy system modeling is the occasional contradictory use of terminology. For instance, the terms of “model” and “model generator” are frequently mixed up. While a “model” in our context is a simplified representation of a real world’s energy system, a “model generator” is a tool that creates a model with predefined properties. Moreover, the term “simulation” is generally used for forecasting a model’s behavior in a future scenario. Since “simulation” also takes place in optimization and back-casting operations, the use of the term is indeed misleading. To avoid misunderstandings, “forecasting” should be used instead when describing future scenarios.

In sum, there is a lack of tools for the modeling and optimization of MES in mixed-use districts. This is especially the case when optional criteria such as non-financial optimization and open source availability are required. Such a small number of suitable tools carries the risk that modeling results cannot be compared with each other, and thus errors as well as potential ways to improve the model are difficult to detect. Nevertheless, the availability of at least two tools shows that there is research and development in the area under consideration, although a wider variety would be desirable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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